

CSS Selectors

Target Any HTML Element With Precision

■ CSS SELECTORS

■ LEARNING OUTCOME

By the end of this module you will use every type of CSS selector — element, class, ID, attribute, pseudo-class, pseudo-element, and combinators — and calculate specificity to debug styling conflicts.

■ Skills You Will Gain

- Use element, class, ID, and universal selectors
- Write attribute selectors for precise targeting
- Apply pseudo-classes: :hover, :focus, :nth-child, :not
- Use pseudo-elements: ::before, ::after, ::first-line
- Understand and calculate CSS specificity
- Combine selectors with descendant, child, and sibling combinators

■ CSS selectors are the bridge between HTML and CSS. Master selectors and you can style anything on any page. Selector problems are the root cause of 80% of CSS bugs.

SECTION 1

Basic Selectors

```
/* Universal selector – targets EVERY element */ * { box-sizing: border-box; margin: 0; padding: 0; }
/* Element selector – targets all elements of that type */ p { color: #333; line-height: 1.6; }
h1 { font-size: 2rem; font-weight: 700; }
/* Class selector – targets all elements with that class */
.card { background: white; border-radius: 8px; padding: 24px; }
.btn { display: inline-block; padding: 10px 20px; cursor: pointer; }
/* ID selector – targets ONE unique element per page */
#hero { background: linear-gradient(135deg, #667eea, #764ba2); }
#nav { position: sticky; top: 0; z-index: 100; }
/* Multiple classes on one element */
<div class="card card--featured card--large">...</div>
.card--featured { border-left: 4px solid #E44D26; }
.card--large { padding: 48px; }
/* Group selector – apply same rules to multiple selectors */
h1, h2, h3, h4 { font-family: 'Inter', sans-serif; }
.btn-primary, .btn-secondary { border-radius: 6px; font-weight: 600; }
```

SECTION 2

Attribute Selectors

```
/* [attr] – has this attribute */
input[required] { border-color: #E44D26; }
/* required inputs */
a[target] { color: purple; }
/* links with target attr */
/* [attr="val"] – exact value match */
input[type="email"] { background: url(mail-icon.svg) no-repeat right 12px center; }
input[type="password"] { letter-spacing: 0.1em; }
a[rel="noopener"] { text-decoration: underline dotted; }
/* [attr^="val"] – value STARTS WITH */
a[href^="https"] { color: green; }
/* secure links */
a[href^="mailto"] { color: #E44D26; }
/* email links */
a[href^="tel"] { color: blue; }
/* phone links */
/* [attr$="val"] – value ENDS WITH */
a[href$=".pdf"] { padding-right: 20px; background: url(pdf-icon.svg) no-repeat right; }
a[href$=".zip"] { color: orange; }
/* download links */
/* [attr*="val"] – value CONTAINS */
a[href*="github"] { color: #333; }
/* github links */
img[src*="thumb"] { width: 80px; }
/* thumbnail images */
/* [attr~="val"] – value in whitespace-separated list */
[class~="highlight"] { background: yellow; }
/* class="a highlight b" */
/* Case insensitive */
[type="EMAIL" i] { border: 2px solid blue; }
```

SECTION 3

Pseudo-Classes

Pseudo-class	When It Applies
:hover	Mouse pointer is over the element
:focus	Element has keyboard focus (tab, click)
:active	Element is being clicked (mousedown to mouseup)
:visited	Link that has been visited before
:first-child	Element that is the first child of its parent
:last-child	Element that is the last child of its parent
:nth-child(n)	nth element — :nth-child(2) = 2nd, :nth-child(even) = all even
:nth-child(2n+1)	Odd-numbered children (1, 3, 5...)
:not(selector)	Matches elements that do NOT match the selector
:checked	Checked checkboxes or radio buttons

:disabled	Disabled form inputs
:placeholder-shown	Input that currently shows placeholder text
:empty	Element with no children or text content
:root	The html element — use for CSS custom properties

```
/* Navigation active states */ nav a:hover { color: #E44D26; text-decoration: underline; } nav
a:focus { outline: 2px solid #E44D26; outline-offset: 2px; } nav a.active { font-weight: 700;
color: #E44D26; } /* Table zebra striping */ tr:nth-child(even) { background: #f9f9f9; } tr:hover {
background: #fff3f0; } /* Style all inputs EXCEPT submit button */
input:not([type="submit"]):not([type="checkbox"]) { border: 1px solid #ccc; border-radius: 4px;
padding: 8px 12px; } input:focus { border-color: #E44D26; outline: none; box-shadow: 0 0 3px
rgba(228,77,38,.2); } /* Only first list item gets top border */ li:first-child { border-top: none;
} li:not(:last-child) { border-bottom: 1px solid #eee; }
```

SECTION 4

Pseudo-Elements

Pseudo-element	What It Creates
::before	Inserts generated content BEFORE element's content. Needs content: property.
::after	Inserts generated content AFTER element's content. Needs content: property.
::first-line	Targets only the first rendered line of a block element
::first-letter	Targets only the first letter — for drop caps
::placeholder	Styles the placeholder text inside inputs
::selection	Styles text the user has highlighted with cursor
::marker	Styles the bullet point or number in list items

```
/* Required field asterisk — no extra HTML needed */ label.required::after { content: " *"; color:
#E44D26; font-weight: bold; } /* External link icon */ a[href^="http"]::after { content: " ■";
font-size: 0.75em; color: #999; } /* Clearfix hack (modern: use flexbox/grid instead) */
.container::after { content: ""; display: table; clear: both; } /* Decorative background element */
.hero::before { content: ""; /* required, even if empty */ position: absolute; inset: 0;
background: linear-gradient(135deg, rgba(228,77,38,.8), transparent); pointer-events: none; } /*
Custom quote marks */ blockquote::before { content: "■C"; font-size: 4rem; color: #E44D26; }
blockquote::after { content: "■D"; } /* Style placeholder text */ input::placeholder { color: #999;
font-style: italic; } /* Custom text selection colour */ ::selection { background: #E44D26; color:
white; }
```

SECTION 5

Combinators and Specificity

Combinator	Syntax	Selects
Descendant	A B	All B that are ANYWHERE inside A (any depth)
Child	A > B	Only B that are DIRECT children of A
Adjacent Sibling	A + B	Only B immediately AFTER A (same parent)

General Sibling	A ~ B	All B that come AFTER A (same parent)
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CSS Specificity Calculation

Specificity is a score that determines which CSS rule wins when multiple rules target the same element. Calculate as (a, b, c) where: a = number of ID selectors (weight: 100) b = number of class, attribute, pseudo-class selectors (weight: 10) c = number of element, pseudo-element selectors (weight: 1) Examples: div = (0,0,1) = 1 .card = (0,1,0) = 10 #hero = (1,0,0) = 100 #hero .card p = (1,1,1) = 111 style='...' (inline) = (1,0,0,0) = 1000 — overrides everything !important = Nuclear option — NEVER use except for utilities

■ PRACTICE Q & A

Q1. What is the difference between .card p and .card > p?

A: .card p selects ALL p elements anywhere inside .card (descendants). .card > p selects ONLY p elements that are direct children of .card. The child combinator (>) is more specific and performant.

Q2. Why should you almost never use !important?

A: !important overrides the entire specificity system. Once you start using it, you need more !important to override previous ones — escalating into an unmaintainable mess. Use higher specificity selectors instead.

Q3. What does :nth-child(3n+1) select?

A: Elements at positions 1, 4, 7, 10... (every 3rd starting from position 1). The formula is An+B where A is the cycle size and B is the offset.

Q4. What is the difference between ::before and :before (single colon)?

A: CSS3 uses double colon (::) for pseudo-elements to distinguish from pseudo-classes (:). Single colon (:before) is CSS2 syntax — still works in browsers but double colon is the modern standard. Always use ::.

Q5. How do you calculate the specificity of #header nav a:hover?

A: #header = (1,0,0), nav = (0,0,1), a = (0,0,1), :hover = (0,1,0). Total = (1,1,2) = 112. This would override any rule with specificity below 112.

■ CSS Selectors Cheat Sheet	
*	Universal – every element
element	Element type selector
.class	Class selector – reusable
#id	ID selector – unique per page
a[href^='https']	Attribute starts with
a[href\$='.pdf']	Attribute ends with
a[href*='git']	Attribute contains
:hover / :focus	Mouse over / keyboard focus
:nth-child(2n)	Every even child element
:not(.hidden)	Exclude matching elements
::before / ::after	Generated content pseudo-elements
A > B	Direct child only